

# PROBABILITY THEORY & MATHEMATICAL STATISTICS

EG-Baccalaureate Advanced STEM Track Handbook

ACADEMIC YEAR: 2025 / 2026 | SENIOR YEAR (EG-BACCALAUREATE ADVANCED DIPLOMA)

## CURRICULUM OVERVIEW & AUTHORIZED SCOPE

Comprehensive guide to modern probability and mathematical statistics: Bayes rule, continuous random variables, joint densities, sample estimation, confidence intervals, and hypothesis test p-values.

### CORE CURRICULUM HIGHLIGHTS:

- Bayes Theorem in sequential decision trees and conditional distributions
- Continuous PDFs, CDFs, and expectation integrals
- Central Limit Theorem application to large sample distributions
- Confidence intervals and two-tailed Z and t hypothesis tests

### APPROVED CHAPTER MODULES:

- |                                                                        |                            |
|------------------------------------------------------------------------|----------------------------|
| - Discrete Probability, Bayes Theorem & Chebyshev Bound                | pp. 1 104 (4 core units)   |
| - Continuous Distributions, Central Limit Theorem & Hypothesis Testing | pp. 105 212 (4 core units) |

## OFFICIAL PUBLICATION METADATA & VERIFICATION

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# MINISTERIAL PREFACE & STATUTORY FRAMEWORK

## MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

### 1. Pedagogical Scope & National Objectives

The Pure Mathematics: Algebra & Solid Geometry curriculum for Grade 12 develops rigorous analytic and algebraic reasoning. It unites discrete combinatorics, complex analysis in polar and exponential forms, linear systems with matrix algebra, and three-dimensional spatial Euclidean geometry. This syllabus equips students for higher engineering, physics, and computer science degrees.

### 2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

|                                                                             |                                                                                       |                                                                                        |
|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <b>30%</b><br><b>Knowledge &amp; Recall</b><br>Definitions, laws & formulas | <b>40%</b><br><b>Understanding &amp; Application</b><br>Calculations, proofs & models | <b>30%</b><br><b>Higher Order Thinking (HOTS)</b><br>Analysis, synthesis & non-routine |
|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|

### 3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

### 4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal: <https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

# SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 120 Minutes (2 Hours) | Total Paper Marks: 30 Marks

| Chapter / Modular Title                       | Weeks | Marks | Weight | Curricular Focus                                         |
|-----------------------------------------------|-------|-------|--------|----------------------------------------------------------|
| Permutations, Combinations & Binomial Theorem | 5 wks | 7 pts | 23.3%  | Combinatorics, Binomial Expansion, General Term          |
| Complex Numbers & De Moivre Theorem           | 5 wks | 7 pts | 23.3%  | Argand Plane, Polar/Euler Form, nth Roots of Unity       |
| Matrices, Determinants & Linear Systems       | 4 wks | 6 pts | 20.0%  | Rank, Inverse Matrix, Cramer Rule, Gaussian              |
| 3D Cartesian Coordinates & Vectors            | 4 wks | 5 pts | 16.7%  | Dot/Cross Product, Sphere Equation, Direction Cosines    |
| Lines and Planes in 3D Space                  | 4 wks | 5 pts | 16.7%  | Vector/Parametric/Cartesian Equations, Angles, Distances |

## INTERDISCIPLINARY COGNITIVE BRIDGES

|                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Physics (Mechanics &amp; Electromagnetism)</b></p> <p>Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.</p> |
| <p><b>Engineering Sciences (Civil, Mechanical, Electrical)</b></p> <p>Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.</p>               |
| <p><b>Computer Science &amp; Algorithmics</b></p> <p>Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.</p>      |

# CHAPTER SPECIFICATION: CHAPTER 1: PERMUTATIONS, COMBINATIONS & BINOMIAL THEOREM

## A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Master fundamental counting principle with and without replacement, ordered and unordered collections.
- [+] Derive and manipulate permutations  $nPr$  and combinations  $nCr$  identities:  $nCr = nC(n-r)$  and  $nCr + nC(r-1) = (n+1)Cr$ .
- [+] Expand  $(a + b)^n$  using Binomial Theorem, identify the general term  $T_{r+1} = nCr \cdot a^{n-r} \cdot b^r$ , find middle terms and greatest coefficient.
- [+] Solve advanced systems involving ratios of consecutive terms:  $T_{r+1} / T_r = ((n - r + 1) / r) \cdot (b / a)$ .

## B. Statutory Theorems & Analytical Principles

**Fundamental Counting Principle:** If task A has  $m$  ways and task B has  $n$  ways, sequential completion has  $m \cdot n$  ways.

**Binomial Theorem:**  $(a + b)^n = \sum_{r=0}^n [ nCr \cdot a^{n-r} \cdot b^r ]$  for positive integer  $n$ .

**Pascal Combinatorial Recurrence:**  $nCr + nC(r-1) = (n+1)Cr$ .

**Consecutive Term Ratio Theorem:**  $T_{r+1} / T_r = ((n - r + 1) / r) \cdot (Term_2 / Term_1)$ .

## C. Standard Mathematical Expressions & Operational Formulas

- >  $nPr = n! / (n - r)!$
- >  $nCr = n! / [ r! (n - r)! ] = nPr / r!$
- >  $nCr / nC(r-1) = (n - r + 1) / r$
- >  $T_{r+1} = nCr \cdot (x)^{n-r} \cdot (y)^r$  in expansion of  $(x + y)^n$
- > Sum of coefficients in  $(a x + b)^n = (a + b)^n$  [Substitute  $x = 1$ ]

## D. National Examination Pitfalls & Common Student Errors

[!] Confusing order significance: using permutations where combinations are needed in geometric configurations (diagonals, triangles).

[!] Neglecting negative signs inside terms when computing  $T_{r+1}$  in expansions like  $(2x - 3/x)^n$ .

[!] Failing to verify integer constraints on  $r$  when seeking the term independent of  $x$ .

## CHAPTER SPECIFICATION: CHAPTER 2: COMPLEX NUMBERS & DE MOIVRE THEOREM

### A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Represent complex numbers in Cartesian  $z = x + i y$ , trigonometric/polar  $z = r (\cos \theta + i \sin \theta)$ , and Euler exponential  $z = r e^{i \theta}$  forms.
- [+] Apply De Moivre theorem for integer and rational exponents:  $[r (\cos \theta + i \sin \theta)]^n = r^n (\cos n \theta + i \sin n \theta)$ .
- [+] Extract all  $n$ -th roots of a complex number evenly distributed on the circle of radius  $r^{1/n}$ .
- [+] Operate with the cubic roots of unity  $1, \omega, \omega^2$  and their algebraic properties ( $1 + \omega + \omega^2 = 0, \omega^3 = 1$ ).

### B. Statutory Theorems & Analytical Principles

**De Moivre Theorem:**  $(\cos \theta + i \sin \theta)^n = \cos(n \theta) + i \sin(n \theta)$  for all rational  $n$ .

**Euler Identity:**  $e^{i \theta} = \cos \theta + i \sin \theta, e^{i \pi} + 1 = 0$ .

**Roots of Unity Theorem:** The  $n$  roots of 1 form vertices of a regular  $n$ -gon centered at the origin on the unit circle.

**Properties of Omega:**  $\omega^2 + \omega + 1 = 0, \omega - \omega^2 = \pm i \sqrt{3}, \omega^{3k} = 1$ .

### C. Standard Mathematical Expressions & Operational Formulas

- > Modulus:  $r = |z| = \sqrt{x^2 + y^2}$ , Principal argument:  $-\pi < \theta \leq \pi$
- > Multiplication:  $z_1 * z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$
- > Division:  $z_1 / z_2 = (r_1 / r_2) [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$
- > Roots:  $z^{1/n} = r^{1/n} [\cos((\theta + 2 k \pi)/n) + i \sin((\theta + 2 k \pi)/n)], k = 0, 1, \dots, n-1$
- > Omega identities:  $1 + \omega = -\omega^2, 1 + \omega^2 = -\omega, \omega + \omega^2 = -1$

### D. National Examination Pitfalls & Common Student Errors

[!] Calculating  $\arg(z)$  incorrectly in quadrants 2, 3, and 4 (must maintain  $-180 < \theta \leq 180$  degrees).

[!] Missing the imaginary unit  $i$  in roots or confusing  $(\omega - \omega^2)^2 = -3$ .

[!] Applying De Moivre to non-standard trigonometric forms like  $(\sin \theta + i \cos \theta)$  without shifting by  $(\pi/2 - \theta)$ .

## CHAPTER SPECIFICATION: CHAPTER 3: MATRICES, DETERMINANTS & LINEAR SYSTEMS

### A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Evaluate 2x2 and 3x3 determinants using cofactor expansion and Laplace development.
- [+] Apply triangular matrix determinant theorem and properties of row operations.
- [+] Find the adjugate matrix and multiplicative inverse  $A^{-1} = (1 / \det(A)) * \text{adj}(A)$ .
- [+] Determine matrix rank  $\text{Rank}(A)$  and augmented matrix rank  $\text{Rank}(A|B)$  to classify system solvability (Rouche-Capelli Theorem).

### B. Statutory Theorems & Analytical Principles

**Determinant Transpose Theorem:**  $\det(A^T) = \det(A)$ .

**Product Theorem:**  $\det(A B) = \det(A) * \det(B)$ ;  $\det(k A) = k^n \det(A)$  for  $n \times n$  matrix.

**Invertibility Theorem:** Square matrix  $A$  is invertible if and only if  $\det(A) \neq 0$ .

**Rouche-Capelli Theorem:** Consistent system iff  $\text{Rank}(A) = \text{Rank}(A|B)$ . Unique solution if  $\text{rank} = n$ ; infinite solutions if  $\text{rank} < n$ .

### C. Standard Mathematical Expressions & Operational Formulas

- > Inverse matrix:  $A^{-1} = (1 / |A|) * \text{Adj}(A)$
- > System solution:  $X = A^{-1} * B$
- > Cramer Rule:  $x_i = \det(A_i) / \det(A)$
- > Homogeneous system: Has non-trivial solutions iff  $\det(A) = 0$

### D. National Examination Pitfalls & Common Student Errors

[!] Confusing matrix multiplication properties:  $AB \neq BA$  in general.

[!] Sign errors when computing cofactor matrix (remember checkerboard pattern + - + -).

[!] Forgetting that row operations change determinant values (swapping rows multiplies det by -1).

# OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

| COMBINATORICS & BINOMIAL THEOREM                |
|-------------------------------------------------|
| $nPr = n! / (n - r)!$                           |
| $nCr = n! / [r! (n - r)!]$                      |
| $nCr = nC(n-r)$                                 |
| $nCr + nC(r-1) = (n+1)Cr$                       |
| $nCr / nC(r-1) = (n - r + 1) / r$               |
| $T_{(r+1)} = nCr * a^{(n-r)} * b^r$             |
| $T_{(r+1)} / T_r = ((n - r + 1) / r) * (b / a)$ |

| MATRICES & DETERMINANTS                         |
|-------------------------------------------------|
| $A^{(-1)} = (1 /  A ) * Adj(A)$                 |
| $ A^T  =  A ,  A B  =  A   B ,  k A  = k^n  A $ |
| $X = A^{(-1)} B$ (Matrix Equation)              |
| $Rank(A) \leq \min(m, n)$                       |

| COMPLEX NUMBERS & DE MOIVRE                                                       |
|-----------------------------------------------------------------------------------|
| $z = x + i y = r (\cos \theta + i \sin \theta) = r e^{(i \theta)}$                |
| $r = \sqrt{x^2 + y^2}, \tan \theta = y / x$                                       |
| $z^n = r^n [\cos(n \theta) + i \sin(n \theta)]$                                   |
| $z^{(1/n)} = r^{(1/n)} [\cos((\theta + 2k \pi)/n) + i \sin((\theta + 2k \pi)/n)]$ |
| $\omega^3 = 1, 1 + \omega + \omega^2 = 0$                                         |
| $\omega - \omega^2 = \pm i \sqrt{3}$                                              |

| VECTORS & 3D ANALYTICAL GEOMETRY                                                               |
|------------------------------------------------------------------------------------------------|
| $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$                                             |
| $u \cdot v =  u   v  \cos \theta = u_1 v_1 + u_2 v_2 + u_3 v_3$                                |
| $u \times v =   \begin{matrix} i & j & k \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{matrix}  $ |
| $d =  a x_0 + b y_0 + c z_0 + d  / \sqrt{a^2 + b^2 + c^2}$                                     |
| $\cos \theta =  n_1 \cdot n_2  / ( n_1   n_2 )$                                                |
| $\sin \theta =  d \cdot n  / ( d   n )$                                                        |

